

Bihar Engineering University, Patna**B.Tech 2nd Semester Examination, 2026****Course: B.Tech****Code: 100215****Subject: Engineering Chemistry****Time: 03 Hours****Full Marks: 70****Instructions:-**

- (i) The marks are indicated in the right-hand margin.
- (ii) There are **NINE** questions in this paper.
- (iii) Attempt **FIVE** questions in all.
- (iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only):**[2 x 7 = 14]**

- (a) The shape of the ammonia (NH_3) molecule according to VSEPR theory is?
 - (i) Trigonal planar
 - (ii) Linear
 - (iii) Trigonal pyramidal
 - (iv) Tetrahedral
- (b) Which of the following metal carbonyls obeys the 18-electron rule?
 - (i) $\text{Fe}(\text{CO})_5$
 - (ii) $\text{Ni}(\text{CO})_4$
 - (iii) $\text{Cr}(\text{CO})_4$
 - (iv) $\text{Mn}(\text{CO})_2$
- (c) UV-Visible spectroscopy mainly involves transitions of
 - (i) Nuclear spins
 - (ii) Core electrons only
 - (iii) Protons only
 - (iv) Valence electrons
- (d) The resonance frequency, in NMR spectroscopy depends mainly on
 - (i) Molecular weight
 - (ii) External magnetic field
 - (iii) Density
 - (iv) Melting point
- (e) The Nernst equation is used to determine
 - (i) Rate of reaction
 - (ii) Heat of reaction
 - (iii) Molecular weight
 - (iv) Cell potential under non-standard conditions
- (f) Which electrolyte used in a lead-acid battery
 - (i) NaOH
 - (ii) Dilute H_2SO_4
 - (iii) HCl
 - (iv) KOH
- (g) Temporary hardness of water is mainly due to
 - (i) Calcium chloride
 - (ii) Magnesium sulfate
 - (iii) Sodium chloride
 - (iv) Calcium bicarbonate
- (h) Which process removes hardness by exchanging Ca^{2+} and Mg^{2+} ions with Na^+ ions?
 - (i) Lime-soda process
 - (ii) Zeolite process
 - (iii) Distillation
 - (iv) Reverse osmosis
- (i) Which polymer is an example of a thermoplastic?
 - (i) Bakelite
 - (ii) Melamine
 - (iii) Polyethylene
 - (iv) Epoxy resin
- (j) Silicones are classified as
 - (i) Natural polymers
 - (ii) Condensation monomers
 - (iii) Biopolymers
 - (iv) Inorganic polymers

- Q.2 (a) Discuss the Photoelectric Effect and Heisenberg's Uncertainty Principle with suitable examples. [7]
 (b) Describe the formation of Molecular Orbitals using the Linear Combination of Atomic Orbitals. Draw and explain the molecular orbital diagram of O₂. [7]
- Q.3 (a) Explain the principles of rotational and vibrational spectroscopy and applications of these techniques in structural elucidation of organic molecule. [7]
 (b) Describe the principle, instrumentation, and applications of UV-Visible spectroscopy in chemistry. [7]
- Q.4 (a) Derive the Nernst equation and explain its significance in determining the EMF of electrochemical cells under non-standard conditions. [7]
 (b) Explain the mechanism of corrosion. Discuss different types of corrosion with suitable examples. [7]
- Q.5 (a) What is hardness of water? Explain the types of hardness and describe the estimation of hardness by the EDTA method. [7]
 (b) Discuss in details the Lime-Soda process for the removal of hardness of water. [7]
- Q.6 (a) Differentiate between thermoplastics and thermosetting plastics. Discuss the preparation, properties and applications of commercially important thermoplastics such as Polyethylene, Polyvinyl Chloride and Polystyrene. [7]
 (b) Describe the preparation, properties and applications of Phenol-Formaldehyde resin (Bakelite). [7]
- Q.7 (a) Explain the Diels-Alder cyclization reaction and epoxide ring-opening reaction and their applications in organic synthesis. [7]
 (b) Discuss the synthesis of aspirin from salicylic acid. Explain the reaction mechanism and significance of aspirin as an important pharmaceutical drug. [7]
- Q.8 Write Short note on:
 (a) Chemical Shift in NMR spectroscopy. [7]
 (b) Working and advantages of fuel cells. [7]
- Q.9 Write short note on *any two* of the following: [7 x 2 = 14]
 (a) Bonding metal carbonyls
 (b) Boiler problems such as sludge formation, scale formation, priming and foaming
 (c) Oxidation-reduction reactions in organic chemistry
 (d) Mechanism of free radical polymerization reaction